

# Model Cards and Human Review

Project: Responsible Plant Classifier

## 1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Proxy Features and Fairness

Without opening the old lesson, explain one key idea from Proxy Features and Fairness. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Improve Generalization

Without opening the old lesson, explain one key idea from Improve Generalization. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Overfitting

Without opening the old lesson, explain one key idea from Overfitting. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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## 2. Practise with fading help

Use W2: Check arithmetic for Model cards and review. Record the answer once; the lesson and workbook refer to the same attempt.

**Given inputs · Procedure I chose · Middle steps · Result and check**

**Given inputs:** \_\_\_\_\_

**Procedure I chose:** \_\_\_\_\_

**Middle steps:** \_\_\_\_\_

**Result and check:** \_\_\_\_\_

\_\_\_\_\_

**The step I completed without help, and the hint I needed**

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## 3. Individual reasoning check

Use the worked case for Model cards and review. Proposed explanation: “A final answer is enough”.  
Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

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## 4. Independent transfer case

## Independent case: Apply independently

Review this card: model v 2, 90 percent accurate, works for every school. No dataset size, split or baseline is given. Write a correction request with a bounded replacement claim.

### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## 5. Your learning target

Draft five model-card headings for a classroom leaf classifier. Why must the card state that its labels are classroom categories rather than professional plant diagnosis?

## 6. Solve the given case

A model card says healthy/spotted,80% accuracy only.

Add responsible-use evidence.

Show your trace, calculation or evidence

### Write or draw your reasoning and evidence.

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## 7. Check a changed case

A model card says 90% accuracy but gives no intended use, data conditions or human-review rule. What should be added before classroom use?

My independent answer and reason

**Write or draw your reasoning and evidence.**

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### Week 33 Test and reflect

## 8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

### Test record · My evidence

**Test record:** Changed input or condition

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Expected answer or fallback

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Observed result

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Why the result occurred

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

## 9. Project checkpoint

Complete the Plant Classifier model card and require teacher sign-off before any classroom demonstration.

The project change and evidence I will keep

**Write or draw your reasoning and evidence.**

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## 10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review \_\_\_\_\_ Next step \_\_\_\_\_

## 11. Transfer practice from the original programme

Draft five model-card headings for a classroom leaf classifier. Why must the card state that its labels are classroom categories rather than professional plant diagnosis?

**My independently reasoned answer**

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Record support used: none, hint or teacher explanation.

## 12. W1 Draft a purpose

Write an intended-use and non-use statement for the worked card.

**My reasoning and evidence**

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## 13. W2 Check arithmetic

Verify the worked accuracy, precision and recall.

**My reasoning and evidence**

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## 14. W3 Find missing evidence

A card says fair and reliable but provides only accuracy. Request three additions.

**My reasoning and evidence**

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## 15. W4 Apply independently

Review this card: model v 2, 90 percent accurate, works for every school. No dataset size, split or baseline is given. Write a correction request with a bounded replacement claim.

### My reasoning and evidence

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## 16. W5 Maintain the record

What should change in a card after a threshold revision?

### My reasoning and evidence

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## 17. Feedback, return check and learning record

### Evidence · My record

**Evidence:** Worked alone / hint / model

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** First step I repaired and why

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** New case checked later; date and result

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** What I can now explain without notes

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** One remaining question

**My record:** \_\_\_\_\_

\_\_\_\_\_

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

## 18. Pictorial case practice: Model Cards and Human Review

Write a model card whose performance statement can be traced to a concrete evaluation.

Toy  $v = 3$  classifier scored 16/20 on five unseen plants with four views each, under indoor tablet lighting. Baseline scored 14/20.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.



## Model Cards and Human Review

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Name evidence

Record  $v = 3$ , 20 photo cases, 5 plant identities, group-separated split and indoor tablet conditions.



#### 2. Compare baseline

$v = 3$  accuracy 80%; baseline 70% on the same 20 cases.



#### 3. State limits

Do not claim all cameras, outdoor conditions or disease diagnosis were validated.

### CHANGE ONE THING

The publicity sentence becomes "80% accurate for every plant".

Explain the changed result using the matching lesson.

Write a bounded replacement for the publicity sentence. Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

### My picture-based explanation and changed-case prediction

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Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.